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Title of the Report:
Individual Research Component Feasibility Study

Project Name:
VR & Mobile Application Solutions for Maternal Education

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Executive Summary

This feasibility study evaluates a Virtual Reality (VR)–based scenario learning system designed to enhance midwifery education in Sri Lanka. Current midwifery training relies heavily on classroom instruction and limited clinical exposure, offering few opportunities for students to practice clinical decision-making in realistic yet safe environments. This limitation negatively affects learning effectiveness and student confidence.

The proposed VR system delivers immersive, scenario-based learning experiences covering antenatal, intrapartum, and postnatal care. Using 3D cartoon-style environments, students observe maternal and newborn conditions and respond through interactive questions and multiple-choice decisions. Immediate feedback and guided explanations support active learning, critical thinking, and repeatable practice without risk to real patients.

The feasibility analysis confirms that the project is technically achievable using Unity, OpenXR, and standard VR hardware. Operationally, the system can be easily adopted by midwifery students and educators. Financial requirements are minimal, as the project uses existing tools and simulated data only. Ethical considerations are fully addressed, as no real patient data is collected. Overall, the study confirms that the proposed VR-based learning system is feasible, practical, and suitable for improving midwifery education in Sri Lanka.

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Introduction

Background

This individual research component is part of the broader project titled 'VR & Mobile Application Solutions for Maternal Education'. The VR component focuses on improving midwifery education by introducing immersive, scenario-based learning that supports clinical decision-making.

Midwifery students in Sri Lanka primarily learn through lectures, textbooks, and limited clinical placements. Due to time constraints, patient safety concerns, and limited case exposure, students often lack sufficient opportunities to practice decision-making in real clinical environments. Virtual Reality offers a safe and engaging alternative to address this gap.

Objective of the Feasibility Study

The objective of this feasibility study is to assess whether the proposed VR-based scenario learning system can be successfully developed, implemented, and sustained as part of midwifery education.

Scope of the Feasibility Study

This study evaluates the following feasibility aspects:

- **Technical Feasibility** – suitability of VR tools, software platforms, system performance, and development risks
- **Educational Feasibility** – relevance to midwifery education and alignment with clinical learning outcomes
- **Financial Feasibility** – development, testing, and deployment cost considerations
- **Operational Feasibility** – usability for students and educators, system maintenance, and scalability
- **Ethical Feasibility** – learner safety, data privacy, and use of simulated clinical scenarios

Problem Statement

In Sri Lanka, midwifery education relies heavily on classroom-based instruction and limited clinical exposure, providing insufficient opportunities for students to practice clinical decision-making in realistic yet safe environments. This limitation negatively affects learning effectiveness and student confidence.

Project Description

Proposed Solution

The proposed solution is a VR-based interactive learning system developed using Unity and OpenXR. The system presents 3D cartoon-style maternal health scenarios where students observe patient symptoms and make clinical decisions through in-VR questions and quizzes.

Key Features Implemented

- Scenario-based antenatal, intrapartum, and postnatal simulations
- Interactive decision-making using MCQs
- Instant feedback and guided explanations
- Safe, repeatable learning without real patient involvement
- Progress tracking through scores and responses

Technology Used

- Unity game engine
- OpenXR framework
- C# scripting
- VR headset (Meta Quest / PC-based VR)

Feasibility Analysis

Technical Feasibility

The system is technically feasible using existing VR development tools. The researcher has prior experience with Unity and interactive media development. A functional VR prototype has already been developed.

Market Feasibility

The primary users are midwifery students and educational institutions in Sri Lanka. There is a clear need for immersive learning tools due to limited clinical exposure.

Financial Feasibility

The project requires minimal financial investment. Existing hardware and free or low-cost software tools are used

Operation Feasibility

The system is easy to operate and does not require specialized training. It can be integrated into existing educational settings.

Ethical and Environmental Feasibility

The system uses simulated data only and does not collect personal or patient data. All scenarios are designed to be educational, safe, and ethically compliant.

Risk Analysis

Risk	**Impact**	**Mitigation**
VR performance lag	Medium	Optimized 3D assets, reduced polygon count, and efficient scene loading
User discomfort or motion sickness	Low	Minimal camera movement, smooth transitions, and comfortable VR design

Limited user engagement	Medium	Interactive scenarios, decision-based questions, and instant feedback
Technical compatibility issues	Low	Use of standard VR frameworks (Unity, OpenXR) and cross-device testing
Scenario misinterpretation	Low	Clear visual cues, guided explanations, and structured feedback

Recommendations

It is recommended to proceed with full development and evaluation of the VR learning system. Future work may include additional scenarios and mobile application integration.

Conclusion

This feasibility study confirms that the proposed VR-based scenario learning system is practical, achievable, and beneficial for midwifery education in Sri Lanka.

References

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